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Optimizing Solvent Systems for Effective Lutein Solvation: A Molecular Dynamics Approach Using 2-Methyl Tetrahydrofuran, Deep Eutectic Solvent, Microemulsion, and Hexane

*Ayush S. Gulhane, Rontu Das, Navnath T. Hatvate, and Debashis Kundu**

The critical importance of solvent selection in lutein solvation lies in its significant commercial value and health benefits. This study employs molecular dynamics (MD) simulations to elucidate the atomic-level interactions of lutein with 2-methyl tetrahydrofuran (2-MeTHF), a microemulsion system (Tween 80: n-propanol: isoamyl acetate: water), a deep eutectic solvent (DES) (choline chloride: glucose, 1:3 molar ratio), and hexane. This study addresses the gap in atomic-scale understanding of lutein-solvent interactions by providing a comparative analysis. MD simulations offer comprehensive insights into structural properties, such as radial distribution functions (RDF), hydrogen bonding (H-bonding) dynamics, and transport properties like mean square displacement (MSD) within these solvent systems. The analysis reveals that DES exhibits the highest suitability for lutein solvation among the solvents studied, suggesting its potential to enhance extraction efficiency while significantly reducing environmental impact, which is validated by experimental literature. These findings contribute to developing more efficient, sustainable, and environmentally friendly extraction techniques for lutein and similar bioactive compounds, potentially benefiting the extraction of other xanthophylls and structurally related molecules.

1. Introduction

Lutein ($C_{40}H_{56}O_2$)[1] is a naturally occurring xanthophyll (oxy carotenoid).[2] It is primarily found in high concentrations in marigold flowers (genus Tagetes)[3] as well as in green leafy vegetables like spinach, kale, and broccoli.[4,5] Marigold is native to Mexico and Central America, and is commercially cultivated as an ornamental plant and traded worldwide,[6] and has naturalized themselves in regions including the south-western United States, South America, Africa, China, and India.[2,7–12] Marigold petals predominantly contain lutein in their esterified forms. However, research has shown that small amounts of free lutein exist in untreated marigold extracts.[13] A comprehensive analysis of lutein compounds in marigold flowers without pretreatment, such as saponification, demonstrates the presence of free lutein in untreated marigold flower extracts across various cultivars.[14] Specifically, free lutein concentrations in *Tagetes erecta* L. species are 28.2 μg g^{-1} in Antigua Orange and 13.8 μg g^{-1} in Antigua Yellow. For *T. patula* L. species, the concentrations are 31.3 μg g^{-1} in Durango Red, 17.9 μg g^{-1} in Hero Bee, 27.4 μg g^{-1} in Janie Primrose, 27.5 μg g^{-1} in Safari Scarlet, and 18.1 μg g^{-1} in Safari Yellow, all on an "as-is" basis. These free lutein levels constitute ≈0.5–8% of the total lutein content, and quantitative profiling of carotenoids in ten marigold cultivars further supports these findings.[15]

Lutein is recognized as an essential supplement for preventing ocular diseases, effectively lowering the risk of age-related macular degeneration (AMD) and cataracts.[10,16,17] It serves as a crucial component of the macular pigment in the retina, shielding against retinal damage by filtering harmful short-wavelength visible light and neutralizing oxidative stress from reactive oxygen species.[18,19] Given its vital role in eye health, lutein holds a 23% market share among carotenoids.[20,21] The global lutein market, valued at approximately USD 364.71 million in 2024, is expected to grow at a compound annual growth rate (CAGR) of 6.29%, reaching USD 494.78 million by 2029.[22] This growth is fueled by the rising demand for eye health supplements and the increasing incidence of myopia and various ocular disorders, including AMD, xerophthalmia, cataracts, glaucoma, and inflammatory conditions, in the population.[23] Lutein is known for its diverse biological activities, including antioxidant activities, anti-inflammatory properties, protection against ultraviolet light,[24] prevention of cardiovascular hardening caused by ageing and coronary heart disease,[25–28] and neuroprotective and anticancer properties[29] while holding significant economic value. Efforts by governmental bodies, coupled with a growing consumer incli-

A. S. Gulhane, R. Das, N. T. Hatvate, D. Kundu
Department of Chemical Engineering
Institute of Chemical Technology Marathwada Campus
Jalna, Maharashtra 431203, India
E-mail: d.kundu@marj.ictmumbai.edu.in

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nation toward natural and preventative health measures, significantly contribute to the market's growth.

Lutein extraction from marigolds has become a research focus due to its therapeutic benefits, inability to be synthesized by the human body, and commercial value.[30] Researchers have explored various methods for extracting and optimizing lutein from marigold flowers.[31] Conventional extraction methods are commonly used, such as simple maceration and Soxhlet extraction.[32] In contrast, novel methods like supercritical fluid extraction, enzyme-assisted extraction, deep eutectic solvent (DES) based extraction, microemulsion extraction, and microwave-assisted extraction have been developed.[33–37] During the extraction process, solvation of lutein remains the most common method for solvent extraction due to its efficiency in isolating lutein, especially using single organic solvents like ethanol, dimethyl ether, methanol, ethyl acetate, hexane, tetrahydrofuran, petroleum ether, 2-dichloroethane, acetone, chloroform as well as a mixture of organic solvents like hexane- acetone- toluene, ethanol-water solutions.[8,12,15,38–40] Despite the potential of various solvents for the solvation of lutein, which potentially gives the extraction of lutein from marigold flowers, experimental approaches reveal several critical areas for improvement. Organic solvents, commonly used in extraction processes, are derived from petrochemical sources and pose significant environmental and health risks due to their flammability, volatility, and toxicity.[41,42] Additionally, these solvents often exhibit low solubility for lutein, leading to inefficient extraction processes and the co-extraction of undesirable compounds. Conventional solvent extraction methods have low solubility of lutein, which is also associated with long extraction times, high solvent consumption, and thermal degradation of sensitive carotenoids.[43] However, solvents derived from petrochemicals, such as hexane, are now subjected to stringent regulations imposed by European Directives and the Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH), ensuring their safety and efficacy for use in chemical processes.[39,44]

To address the issues with conventional solvents, the development of greener and more sustainable solvents is necessary to enhance the solubility of lutein. This leads to developing more efficient extraction techniques that minimize environmental impact and improve extraction efficiency. Green solvents from renewable sources significantly reduce waste and simplify recycling and reuse. These solvents are biodegradable, exhibit minimal ecological impact, and have favorable toxicological profiles.[38] There are several greener solvents available, among them one of the potential candidates is 2-methyl tetrahydrofuran (2-MeTHF). The 2-MeTHF solvent has a lower toxicity, is biodegradable, and possesses favorable physicochemical properties such as moderate polarity and high affinity for hydrophobic compounds like lutein. Its hydrophobic nature helps to recover the solvents from aqueous media for reuse purposes.[39] Another greener solvent is the microemulsion-based solvents used for the extraction of lutein. Microemulsion-based solvents are prepared using the hydrophobic and hydrophilic interactions, enhancing lutein's solubility. The microemulsion-based solvent consisting of Tween80: n-propanol: isoamyl acetate: water (36:36:18:10 wt.%) is chosen to create a stable, isotropic phase capable of solubilizing both hydrophilic and lipophilic components. Tween 80 (a nonionic surfactant) and n-propanol (a short-chain alcohol) enhance the microemulsion's solubilization power and biocompatibility, while isoamyl acetate provides a hydrophobic environment favorable for lutein extraction. Water inclusion promotes phase stabilization.[45] Another greener solvent for the extraction process is DES-based solvents. However, the DES is used for several applications such as dehydrogenation,[46,47] metal processing, organic synthesis,[48] extraction,[49] electrochemistry,[50,51] biosensors,[52] and food applications.[53] The DES system, composed of glucose and choline chloride (ChCl) in a 1:3 molar ratio, demonstrates a lutein yield of 368.64 g kg^{-1}.[54] These novel solvents have demonstrated higher solubility of lutein, which leads to higher extraction efficiency than conventional solvents. Although green solvents come at a higher market price than hexane, their improved extraction efficiency and environmental advantages more than compensate for this expense. Nevertheless, acquiring experimental data can be both time-intensive and expensive in evaluating the comparative efficacy of these solvents.

While experimental work has advanced lutein extraction techniques, most studies focus solely on extraction yields and optimization parameters.[55] These methods fail to provide insight into the molecular mechanisms driving lutein-solvent interactions, which are crucial for understanding the extraction process.[56] To fill this gap, this study aims to provide an atomic-level understanding of lutein solvation using molecular dynamics (MD) simulations. Computational methodologies, such as molecular dynamics (MD) simulations, mitigate these shortcomings by offering a practical approach to studying these microscopic processes.[57] Moreover, MD simulations provide dynamic visualizations of systems comprising numerous molecules, revealing insights into behavior on a nanosecond to microsecond timescale. This methodological shift toward MD simulations aligns with the growing recognition of the limitations of solely experimental approaches. It underscores the need for complementary computational techniques to understand extraction processes comprehensively.[58] Numerous computational studies employing MD simulations exploring the structural and transport properties of bioactive compounds derived from natural products interacting with different solvents are extensively documented in the literature.[59] For instance, the extraction of patchouli oil, particularly the interactions between patchoulol and various solvents, concludes that ethanol provides superior solubility, aligning with the experimental yields.[60] A detailed study on the solid-liquid extraction of puerarin, a bioactive component from *Radix Puerariae*,[56] using MD simulations, reveals that the dissolution behaviors vary with ethanol concentration due to hydrogen bonding (H-bonding) interactions between puerarin and the solvents. These studies highlight the significant impact of MD simulations in enhancing the understanding and optimization of the solvation of lutein, which optimizes the extraction processes, thereby improving extraction efficiency and extraction yield.[61] Free lutein, even in low concentration, in untreated extracts provides a basis for studying its interactions with solvents through MD simulations.

In this study, MD simulations elucidate the dynamics of lutein interactions with solvents, including hexane, 2-MeTHF, a microemulsion system, and DES. Despite the absence of thorough comparative studies, these solvents have been selected for this research to identify the most effective option through detailed simulations and analysis. The dynamics reveal insights into inter-

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Table 1. System composition and number of initial molecules involved of dynamic systems.

System name	System abbreviation	Components of system	No. of atoms	No. of molecules	Total no. of atoms	Initial box dimension [nm^3]	Final box dimension [nm^3]	Final density [kg.m^{-3}]
Lutein-2 Methyl tetrahydrofuran	Lutein-2-MeTHF	Lutein	14994	153	30002	50 × 50 × 50	6.67 × 6.67 × 6.67	932.018
		2-Methyl tetrahydrofuran	15008	938				
Lutein-DES	Lutein-DES	Lutein	14994	153	30034	12 × 12 × 12	6.62 × 6.62 × 6.62	1160.25
		Choline chloride	3521	160				
		Glucose	11519	480				
Lutein-Microemulsion	Lutein-Microemulsion	Lutein	14994	153	29948	12 × 12 × 12	6.66 × 6.66 × 6.66	953.493
		Tween 80	4998	49				
		n-propanol	5928	494				
		Isoamyl acetate	2645	115				
		Water	1383	461				
Lutein- Hexane	Lutein- Hexane	Lutein	14994	153	29994	12 × 12 × 12	6.76 × 6.76 × 6.76	818.812
		Hexane	15000	750				

molecular interactions, structural properties such as radial distribution functions (RDF), H-bonding evolution, and transport properties such as mean square displacement (MSD) of lutein within solvent systems. Through this study, it is presumed to gain a deeper understanding of the solvation mechanisms, which will contribute to developing more sustainable and efficient extraction methods for lutein and similar compounds. Future research could apply a similar simulation approach to other xanthophylls and structurally similar molecules.

2. Computational Methodology

2.1. System Details

Initially, the molecular structures of lutein and the solvents, including 2-MeTHF, a DES comprising choline chloride and glucose (1:3 molar ratio), a microemulsion composed of Tween80, n-propanol, isoamyl acetate, and water (36:36:18:10 wt.%), and the conventional solvent hexane are drawn using Avogadro software,[62] which employs the Universal Force Field (UFF) for initial structural refinement.[62] Table S1 (Supporting Information) provides the detailed structure of each solvent component along with the composition ratios of the various lutein-solvent systems under examination.

2.2. System Setup

The lutein and solvent molecules are randomly distributed in a simulation box using PACKMOL package.[63] **Table 1** represents the details of the system and the number of initial molecules involved. The initial simulation box is created with a high volume and low density to guarantee proper organization and prevent structural interference. This spacious configuration helps prevent any initial atomic overlap. Subsequently, the system iteratively reduces the box size until it detects no structural overlaps. This method adequately arranges all atoms, maintaining the integrity of the system's structure.

2.3. Force Field Details

The AMBER 14SB force field is primarily developed for biomolecular systems such as proteins, but it has also been successfully applied to organic liquids and ionic liquids in previous studies.[64–66] Although the OPLS (Optimized Potentials for Liquid Simulations) force field is commonly used for DES-like ChCl/glucose, the present system not only comprises DES components but also includes organic molecules, making AMBER 14SB an appropriate and balanced choice. To validate the suitability of AMBER 14SB for the DES system, the density of the simulated DES system is calculated and compared to the corresponding experimental values. The results, presented in Table S2 (Supporting Information), show good agreement between simulated and experimental densities, supporting the reliability of AMBER 14SB parameters for modelling the DES and organic components in our study. The AMBER14SB force field describes the system's potential energy in these simulations, denoted by the term V_{total}.

$$V_{total} = \sum_{bonds} k_b\left(r - r_0\right)^2 + \sum_{dihedrals} k_\chi\left[1 + \cos(n\chi - \delta_\chi)\right] + \sum_{improper} k_\Psi\left[1 + \cos\left(n\Psi - \delta_\Psi\right)\right] + \sum_{angles} k_\theta\left(\theta - \theta_0\right)^2 + + \sum_{i=1}^{N-1}\sum_{i<j}^{N}\left\{4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}\right\} \quad (1)$$

V_{total} encapsulates the energy components within the system, comprising van der Waals (vdW) interactions, bond stretching, coulombic interactions, angle bending, and proper and improper dihedral torsions. The parameters like the bond stretching constant (k_b), bond length (r), and standard bond length (r_0) influence the bond stretching energy (E_{bond}). The angle bending constant (k_θ), the angle between molecules (θ), and the standard angle (θ_0) dictate the angle bending energy (E_{bend}). The proper dihedral energy ($E_{dihedral}$) is determined by the appropriate angle of the dihedral (χ) between planes, multiplicity (n), phase shift (δ_χ), and dihedral constant ($k\chi$). Similarly, factors like multiplicity fac-

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tor (n), improper dihedral constant (k_ψ), phase shift (δ_ψ), and the improper dihedral angle (ψ) among planes impact the improper dihedral energy ($E_{improper}$). The parameters such as potential well depth (ϵ_{ij}), distance (r) between interacting particles, and the distance at which the zero potential (σ_{ij}) modulates Coulombic and vdW interaction energies. The repulsive term (r_{ij}^{-12}) dominates at short distances, while the attractive term (r_{ij}^{-6}) prevails at longer distances. Additionally, the permittivity of the vacuum is denoted by (ϵ_0), and the charges of two particles are represented by (q_i) and (q_j).

2.4. MD Simulation Framework

The GROMACS 2022.3 package is utilized for all MD simulations[67–69] to evaluate the dynamics of Lutein-Hexane, Lutein-2-MeTHF, Lutein-DES, and Lutein-Microemulsion systems, resulting in simulation trajectories obtained after production simulations. The partial charges for species involved in the simulation are calculated using the Gasteiger method. Energy minimization is performed using the Steepest Descent method to achieve a stable molecular structure, which converges to 1000 kJ mol^{-1} nm over 5×10^6 steps with a step size of 10^{-3} nm. The particle-mesh Ewald (PME) method is utilized,[70] which efficiently computes long-range electrostatic interactions, which is crucial for simulations of charged molecules.[71] The PME order is set to 4, and the Fourier spacing is 0.125 nm. Throughout the equilibrium and production run, the equations of motion are integrated using the Leap-frog integrator[72,73] with a time step of 2 fs. Equilibration unfolds in two stages involving NVT and NPT[74] ensembles. During NVT equilibration, the system is maintained at 300 K for 1 ns. Subsequently, NPT equilibration is conducted at 300 K and 1.013 bar pressure for 19 ns. A velocity-rescale thermostat[75] and the Parrinello-Rahman barostat[76] control temperature and pressure, respectively. A linear constraint solver (LINCS) is used to restrain the H-bonding during the equilibration process.[77] After the NPT equilibration process, total energy, density, temperature, and pressure data are plotted, which are shown in Figure S1 (Supporting Information). It shows the constant end of the dynamics, which suggests the system is well equilibrated. Prior to the production run, system equilibration is validated by analyzing NPT equilibration. The production run extends over 30 ns at 300 Kelvin and 1.013 bar pressure, with coordinates and rates recorded every 10 ps for detailed analysis. After the production run, total energy, density, and temperature are plotted, which are shown in Figure S2 (Supporting Information). It shows the constant throughout the dynamics, which suggests the system has achieved a steady state. The Visual Molecular Dynamics (VMD) package is used to visualize solute-solvent systems.[78] **Figure 1** illustrates the initial and final configurations of the Lutein-Hexane, Lutein-2-MeTHF, Lutein-DES, and Lutein-Microemulsion systems.

2.5. Analysis of Trajectories

2.5.1. Radial Distribution Function (RDF)

The RDF characterises the probability of finding another atom around a specific atom or the likelihood of locating a particle at a particular distance from the centre of another particle.[50] RDF, denoted as $g(r)$, plays a crucial role in MD simulations, illustrating how particle density changes with distance from a reference particle. The mathematical equation of RDF is as follows:

$$g(r) = \frac{1}{N}\left(\frac{V}{4\pi r^2 dr}\right)\sum_{i=1}^{N}\sum_{i\neq j}^{N}\delta\left(|r_i - r_j| - r\right) \quad (2)$$

In Equation (2), N denotes the total number of atoms in the system, while V represents the volume of the simulation box. The variable r indicates the distance between the particles in the system and the reference particle. Additionally, the Dirac delta function, denoted by $\delta(z)$, equals 1 when $z = 0$ and otherwise remains 0. The symbol dr represents the width of the radial bin. The formula for the distance between atoms i and j is $|r_i - r_j|$.

2.5.2. Coordination Number (CN)

The CN measures the number of atoms or molecules in the immediate vicinity of a reference atom or molecule within a specified cutoff distance, providing crucial insights into the localized structure and bonding characteristics of the system. A higher CN indicates tightly packed atoms or molecules with strong interactions, while a lower CN suggests a more porous structure with weaker interactions. The CN is mathematically expressed as:

$$CN(r) = 4\pi\rho N\int_0^r g(r)\, r^2 dr \quad (3)$$

In Equation (3), the RDF, denoted as $g(r)$, is integrated across a spherical shell with a radius r centered on an atom to determine the $CN(r)$. The calculation incorporates particle number density (ρ) and the overall number of atoms in the system (N), while the $g(r)$ characterizes particle distribution around the reference atom.

2.5.3. Hydrogen Bonding (H-Bonding)

A H-bonding is an attractive interaction between a hydrogen atom attached to an electronegative donor group and an electron-rich acceptor, either within the same molecule or in a different one.[79] This interaction is crucial for stabilizing molecular structures, influencing solvation energies, solubility, and determining chemical reactivity.[80] In MD simulation, the H-bond is calculated based on the geometric criteria. A hydrogen bond is formed if α_{HDA}, representing the angle between the hydrogen donor, acceptor, and hydrogen atoms, is less than or equal to 30° and if r_{DA}, indicating the distance between the hydrogen donor and acceptor atoms, is less than 0.35 nm.[81] The number of H-bonds can be calculated mathematically using the following equation:

$$C_{HB} = \frac{\langle H(\tau)\, H(\tau + t)\rangle}{\langle H(\tau)\, H(\tau)\rangle} \quad (4)$$

In Equation (4), the number of H-bonds is expressed as C_{HB}, and the hydrogen bond order parameters at times τ and (τ+t) are represented by the variables $H(\tau)$ and $H(\tau+t)$.

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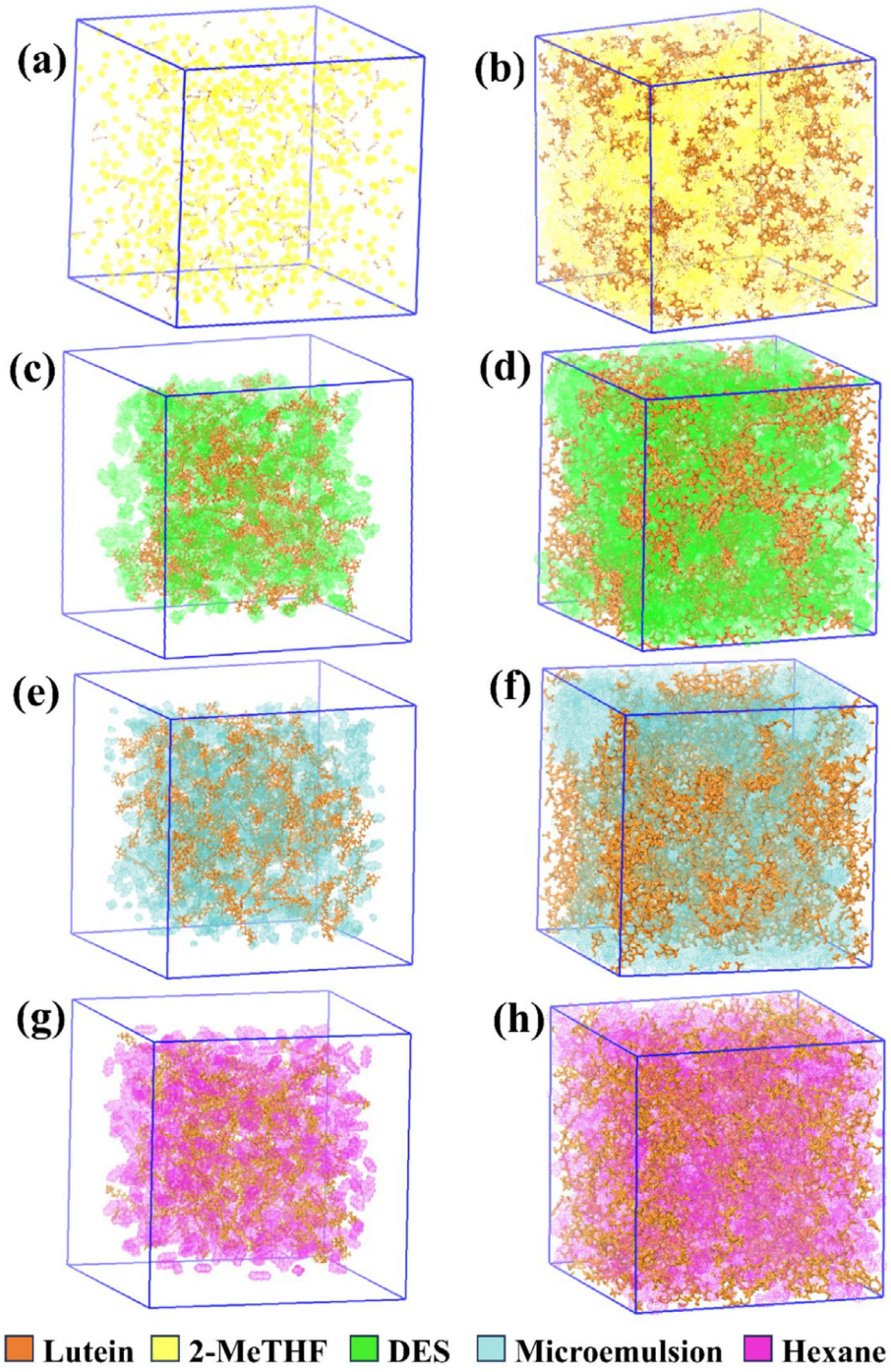


Figure 1. Snapshots of dynamic systems (a), (c), (e), (g) represent the Initial box at 0 ns of Lutein-Hexane, Lutein-2-MeTHF, Lutein-DES, and Lutein-Microemulsion respectively, (b), (d), (f), (h) represents the final box at 30 ns of Lutein-Hexane, Lutein-2-MeTHF, Lutein-DES, and Lutein-Microemulsion respectively.

2.5.4. Mean Square Displacement (MSD)

The MSD characterizes the diffusion and mobility of particles over time. MSD quantifies the average squared distance that a particle moves from its initial position at time *t*. Mathematically, MSD is defined as:

$$MSD(t) = \left\langle \left(r_i(t) - r_i(0)\right)^2 \right\rangle = \frac{1}{N}\sum_{i=1}^{N}\left[\left(r_i(t) - r_i(0)\right)^2\right] \quad (5)$$

where r_i (t) is the position of atom i at time t; $r_i(0)$ denotes the atom's starting position, and the angular brackets (⟨ ⟩) denote an ensemble average over all particles. The self-diffusion coefficient

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(D) is determined using the Einstein formula as represented in Equation (6).[82–84]

$$D = \frac{1}{6}\lim_{t\to\infty}\frac{d\left(MSD\left(t\right)\right)}{dt} = \frac{1}{6N}\lim_{t\to\infty}\frac{d}{dt}\sum_{i=1}^{N}\left[\left(\left(r_i\left(t\right)-r_i\left(0\right)\right)^2\right]\right. \quad (6)$$

The β value is a critical parameter in understanding the diffusive behavior of particles in molecular dynamics simulations. It is derived from the MSD data, providing a quantitative measure of the type of molecular diffusion occurring in the system, such as diffusive ($\beta \approx 1$), subdiffusive (β <1), and superdiffusive ($\beta >$ 1). The mathematical expression is given by Equation (7).

$$\beta = \frac{\ln\left(\frac{MSD\left(t_2\right)}{MSD\left(t_1\right)}\right)}{\ln\left(\frac{t_2}{t_1}\right)} \quad (7)$$

3. Results and Discussion

3.1. Evolution of H-Bonding in the Lutein-Solvent System

Figure 2 illustrates the H-bonding interactions in the Lutein-2-MeTHF, Lutein-DES, and Lutein-Microemulsion systems. The hydroxyl moieties, oxolane (cyclic ethers), and esters act as H-bond donors and acceptors. In Figure 2a, the number of H-bonds between 2-MeTHF and lutein remains relatively stable, fluctuating between 25 and 45 over the 30 ns period, with an average of 33 H-bonds.

The H-bonding interactions in the Lutein-DES system are separately determined with the hydrogen bond acceptor (choline chloride) and hydrogen bond donor (glucose). Figure 2b highlights a fluctuation in the number of H-bonds between glucose and lutein, which oscillates between 100 and 130, with an average of 117 H-bonds. Figure 2c shows that the number of H-bonds between ChCl and lutein in the Lutein-DES system ranges from 1 to 7, with an average of 4 H-bonds. This significant difference in H-bonding can be attributed to the functional groups, mobility, and accessibility of glucose molecules. Glucose possesses multiple hydroxyl groups, which serve as both hydrogen bond donors and acceptors, facilitating a higher degree of H-bonding with lutein than choline chloride. Moreover, the 1:3 molar ratio of choline chloride to glucose molecules in the DES composition further provides more opportunities for glucose to form H-bonding interactions with lutein as suggested by the previous experimental research.[54]

Figure 2d–g represents the H-bond interactions in the Lutein-Microemulsion system. Figure 2d shows the number of H-bonds between tween 80 and lutein, fluctuating between 5 and 20 over the 30 ns, with an average of 13 H-bonds. Figure 2e shows that n-propanol exhibits a stronger interaction with lutein than tween 80, with H-bonds ranging from 5 to 25 and an average of 19 H-bonds. Figure 2f shows the number of H-bonds between isoamyl acetate and lutein, fluctuating between 3 and 9, with an average of 6 H-bonds. However, Figure 2g indicates that the H-bonds between water and lutein fluctuate between 5 and 20 over 30 ns, with an average of 11 H-bonds and a noticeable decline between 10 to 25 ns. This decline is attributed to the competitive interactions within the microemulsion system. The higher affinity of lutein for other components like n-propanol and tween 80 reduces the availability of H-bonding sites for water. Additionally, the lower weight percentage of water (10%) than other components (36% for tween 80 and isoamyl acetate and 18% for n-propanol) further limits its interaction with lutein. However, the inclination in the trend of H-bonding is seen after 25 ns, which suggests restructuring and self-assembly of the molecules. The average number of hydrogen bonds for the overall Lutein-Microemulsion system is 121, highlighting the combined contribution of each component. This comprehensive analysis reveals that n-propanol and tween 80 are the primary contributors to the H-bonding network in the microemulsion system. In contrast, isoamyl acetate and water contribute less to the overall interaction. The H-bonding interaction with lutein in hexane is also analyzed, which shows negligible H-bonding with lutein. This behavior is consistent with the typical characteristics of nonpolar solvents. It suggests that weak non-bonded interactions, such as vdW and electrostatic forces, govern the extraction process in conventional solvents like hexane. The average number of hydrogen bonds per frame for lutein with hexane, 2-MeTHF, microemulsion, and DES is 0, 33, 48, and 121, respectively. The order of H-bonding is Lutein-Hexane < Lutein-2-MeTHF < Lutein-Microemulsion < Lutein-DES. However, the total number of molecules is different, but the number of solvent atoms is nearly the same, as shown in Table 1. The DES system shows higher hydrogen bonds, which is due to a higher number of hydroxyl groups present in the glucose system. The hydroxyl group of glucose acts as a hydrogen bond donor and acceptor. The experimental result also suggest that the DES system has the higher experimental yield which also validated by the H-bonding results.[39,45,54]

3.2. Structural Orientation and Association of Lutein with Solvent

The structural orientation of various functional moieties of lutein in different solvents is analyzed with the help of RDF, and the packing of moieties within the designated solvation shells is represented through the *CN*. **Figure 3** presents the RDF of the Lutein-2-MeTHF system between the oxygen atom of the oxolane moiety of 2-MeTHF and the hydrogen atom of lutein (denoted by a1···a2 in Figure 3). The first solvation shell at 3.35 Å, with a *CN* of 0.05, indicates very limited solvent accessibility. The first solvation shell is observed in the H-bonding interaction regions. The second solvation shell at 6.95 Å ($CN = 0.65$) suggests a gradual increase in solvent molecules around lutein, likely driven by vdW and electrostatic forces. The low *CN* values imply that 2-MeTHF molecules are less densely packed around lutein, limiting close solvent access. Although 2-MeTHF exhibits weak H-bonding according to the RDF data, its moderate polarity may still contribute to experimental extraction efficiency beyond H-bonding through favorable dispersion and dipole–induced dipole interactions.

Figure 4 represents the RDF of various functional moieties of the Lutein-DES system to determine structural orientation throughout the simulation. Figure 4a shows the RDF curve between the chlorine atom of ChCl and the hydrogen atom of the hydroxyl group in glucose (denoted by b1···b2 in Figure 4a), re-

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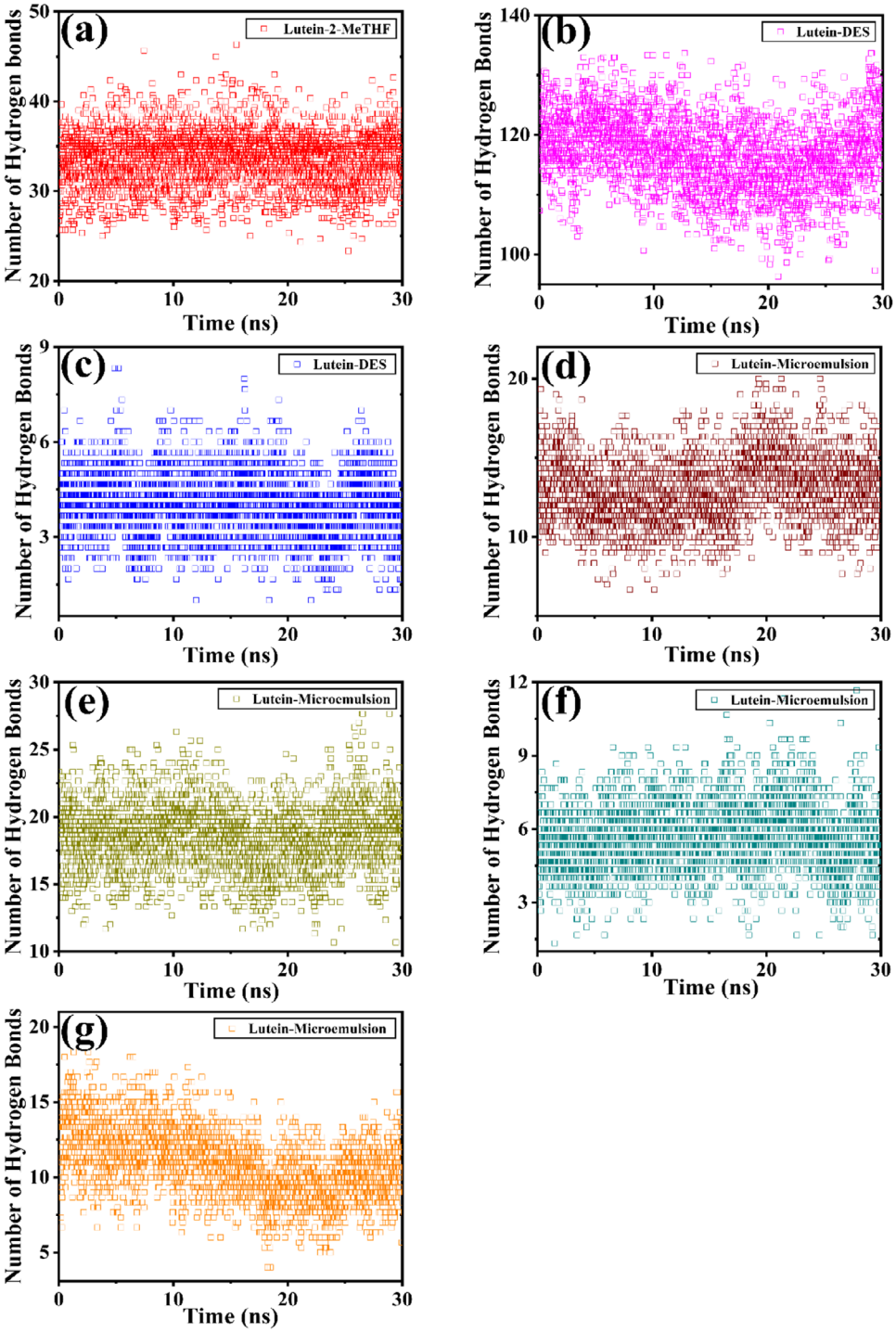


Figure 2. Number of hydrogen bonds between different solvent and lutein systems: a) Lutein-2-MeTHF, b) Lutein-DES (glucose and lutein), c) Lutein-DES (choline chloride and lutein), d) Lutein-Microemulsion (tween 80 and lutein), e) Lutein-Microemulsion (n-propanol and lutein), f) Lutein-Microemulsion (isoamyl acetate and lutein), and g) Lutein-Microemulsion (water and lutein).

vealing significant interactions. The first solvation shell at 3.35 Å, with a *CN* of 0.91, indicates substantial H-bonding and limited immediate solvent accessibility. However, the interaction between ChCl and glucose competes with and weakens the direct H-bonding between lutein and DES components. This is further supported by Figure 4b, where the RDF between the chloride ion of HBA and the hydrogen atom of the hydroxyl group of lutein (denoted by b1···b3) shows a first solvation shell at 3.55 Å with a *CN* of 0.59, suggesting limited solvent accessibility of lutein. Additionally, Figure 4c shows the RDF curve between the hete-

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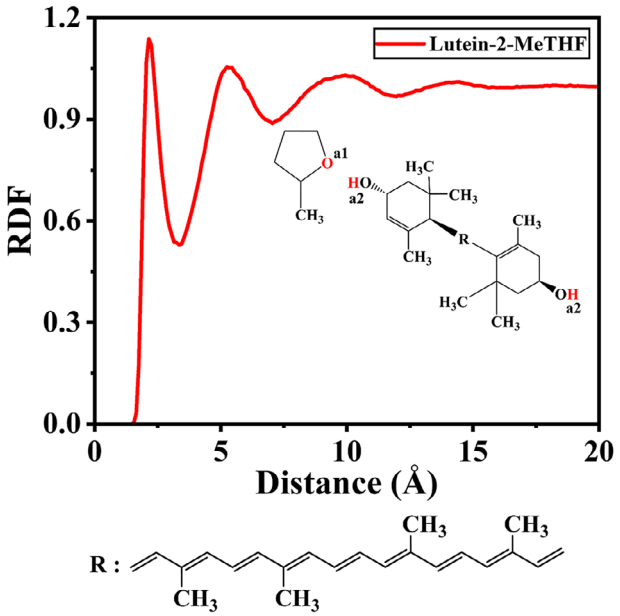


Figure 3. RDF graph of interaction between O (oxolane moiety of 2-Mthf)-H (hydroxyl moiety of lutein).

rocyclic oxygen atom of glucose and the hydrogen atom of the hydroxyl group of lutein (denoted by b4···b3), where the first solvation shell appears at 6.35 Å with a *CN* of 0.48, indicating that the interaction is electrostatic and limited solvent accessibility of lutein.

The RDF curves in **Figure 5** illustrate the interactions among various functional moieties of microemulsion components with lutein. Figure 5a represents the interaction between the oxygen atom of Tween 80 and the hydrogen atom of the hydroxyl group present in lutein (denoted by c1···c2 in Figure 5a). The first solvation shell at 3.15 Å, with a *CN* of 0.05, indicates limited solvent accessibility and interactions present in H-bonding regions. The second and third solvation shells at 6.95 Å (*CN* = 0.73) and 9.65 Å (*CN* = 1.99) are primarily influenced by vdW and electrostatic interactions, reflecting gradual solvent accumulation around lutein. Figure 5b shows the interaction between the oxygen atom of the hydroxyl group of Tween 80 and the hydrogen atom of the hydroxyl group of lutein (denoted by c3···c2). The first solvation shell at 2.95 Å (*CN* = 0.05) and second at 3.45 Å (*CN* = 0.09) suggest even weaker interactions compared to Figure 5a, implying that the hydroxyl group of Tween 80 may not significantly contribute to lutein extraction or transport within the microemulsion system. Similarly, Figure S3a (Supporting Information) depicts the interaction between the heterocyclic oxygen atom of Tween 80 and the hydrogen atom of the hydroxyl group of lutein (denoted by e1···e2), showing very weak interactions with a first solvation shell at 3.05 Å (*CN* = 0.03).

In contrast, Figure 5c highlights the interaction between the oxygen atom in the ester moiety of isoamyl acetate and the hydrogen atom of the hydroxyl group in lutein (denoted by c4···c2 in Figure 5c), where the first solvation shell at 3.25 Å with a *CN* of 0.58 suggests stronger solvent accessibility and more significant interaction. When compared with RDFs generated for lutein-n-propanol (e2···e3 in Figure S3b, Supporting Information) and lutein-water (e2···e4 in Figure S3c, Supporting Information), lutein-isoamyl acetate shows the closest and more robust interaction. In the microemulsion system, n-propanol and isoamyl acetate likely act as co-solvents, enhancing lutein solvation by facilitating better penetration into the micellar structure, while also stabilizing the system by modulating interfacial properties. The micelle formation, mainly by Tween 80, provides a dynamic environment where lutein can be solubilized in the hydrophobic core or interfacial regions, improving overall solvent accessibility.

Figure 6 presents the RDF between the hydroxyl group of lutein and the hydrogen atoms of hexane. The RDF analysis of lutein in hexane primarily reveals electrostatic and vdW interactions, indicated by a larger first solvation shell at 3.55 Å with an average *CN* of 2.40, and a second solvation shell appearing at 6.25 Å with a *CN* of 23. These *CN* values suggest relatively good solvent accessibility, even though interactions are dominated by weak forces. Hexane molecules, being composed entirely of carbon and hydrogen atoms, are non-polar with no permanent dipole moment; thus, traditional H-bonding is absent. Instead, hexane molecules interact via weak, short-range van der Waals forces, particularly London dispersion forces, which arise from temporary electron fluctuations. These transient dipoles promote attraction among neighboring molecules, leading to structural organization and stability around lutein. Although van der Waals forces are weaker than hydrogen bonds, their cumulative effect enables the formation of stable solvation shells in a non-polar environment, supporting effective lutein interaction.

The RDF analysis provides critical insights into the molecular interactions governing lutein extraction across different solvent systems. In the microemulsion system, the low coordination numbers and broader solvation shells reflect limited H-bonding and solvent accessibility, with isoamyl acetate and n-propanol acting as effective co-solvents and stabilizers within the micellar environment. The structure of the microemulsion enhances the dispersion of lutein but relies primarily on van der Waals and electrostatic interactions. In the hexane system, despite the absence of H-bonding, the cumulative effect of van der Waals forces enables stable solvation shells around lutein, ensuring moderate solvent accessibility. In contrast, the Lutein-DES system demonstrates the highest coordination numbers and distinct solvation shell organization, emphasizing the dominance of strong and stable H-bonding networks. These findings confirm that the nature of solvent–solute interactions, the organization of solvation shells, and the structural environment of the solvent critically influence lutein extraction efficiency. Solvents promoting stronger H-bonding and higher solvent accessibility, such as DES, are more favorable for efficient lutein extraction compared to non-polar or weakly interacting systems

3.3. Mean Square Displacement

To examine the extraction behavior of Lutein in various solvents, the diffusion nature of Lutein in all four solvents is depicted in **Figure 7**. Figure 7a shows the MSD of the hydrogen atom of the hydroxyl moiety of lutein in the Lutein-2-MeTHF system. The MSD curve exhibits the β value of 0.70, indicating sub-diffusive behavior. This sub-diffusive behavior results from the interac-

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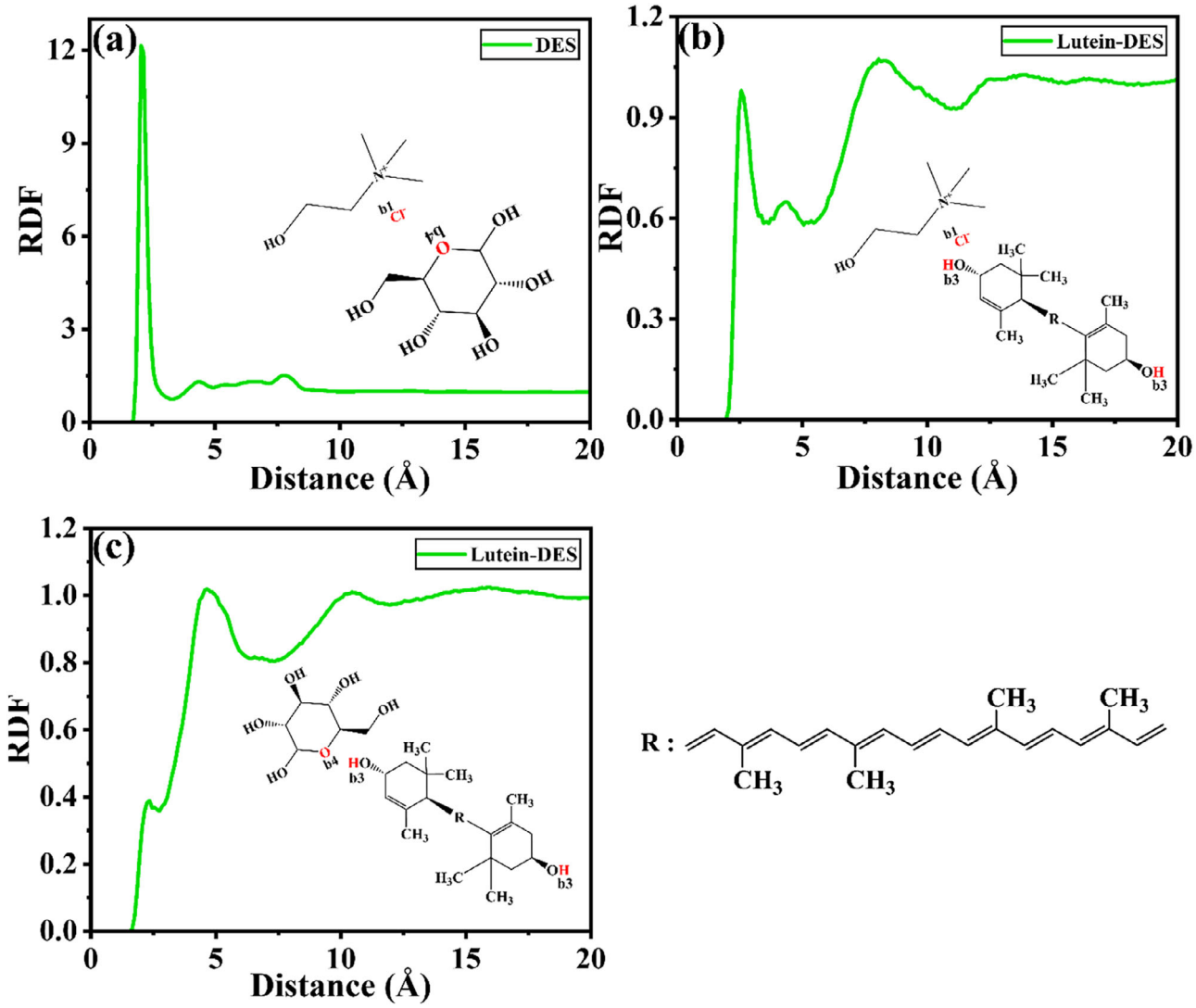


Figure 4. RDF graph of interaction between lutein and components of DES solvent a) Cl (choline chloride)- H (hydroxyl moiety of glucose) (both of DES), b) Cl (choline chloride)- H (hydroxyl moiety of lutein), c) O (hydroxyl moiety of glucose)- H (hydroxyl moiety of lutein).

tions with other solvent moieties that hinder the free diffusion of lutein molecules. Figure 7b illustrates the MSD curve depicting the movement of the hydrogen atom of the hydroxyl moiety of lutein in the Lutein-DES system, with β value of 0.38, indicating strong sub-diffusive behavior influenced by similar H-bonding dynamics. The presence of stronger H-bonding significantly slows down the movement of lutein. Figure 7c represents the MSD curve for the hydrogen atom in the hydroxyl moiety of lutein in the Lutein-Microemulsion system. The curve can be divided into two segments. In the initial segment, lasting until 6.4 ns, the hydrogen atom shows slow movement with a β value of 0.59, indicating subdiffusive behavior due to confinement by strong hydrogen bonds. This behavior is supported by RDF data (Figure 6). In the subsequent segment, the β value marginally increases to 0.64, suggesting a reduction in confinement and a minor increase in diffusivity, likely due to the transient breaking of some hydrogen bonds. Figure 7d shows the MSD curve of the hydrogen atom within the hydroxyl moiety of lutein in the Lutein-Hexane system. The MSD curve shows the β value of 0.73, indicating the presence of sub-diffusion but less restrictive than the other solvents, allowing more freedom of the hydrogen atom in hexane. These four solvents are used for the extraction of lutein. For optimal preservation of the attractive interactions between solute and solvent components, it is crucial to maintain lower molecular movement. Hence, the estimation of the sub-diffusion regime for all the solvents prevailed. The β value in the lutein-DES system is the smallest among the four solvents, indicating the longest residence time for lutein-DES interaction. This supports the experimental evidence that DES is the most effective extractant for lutein.

4. Conclusion

This study comprehensively examines the atomistic interaction of lutein with 2-MeTHF, DES, Microemulsion, and Hexane through MD simulations. The H-bonding analysis reveals that

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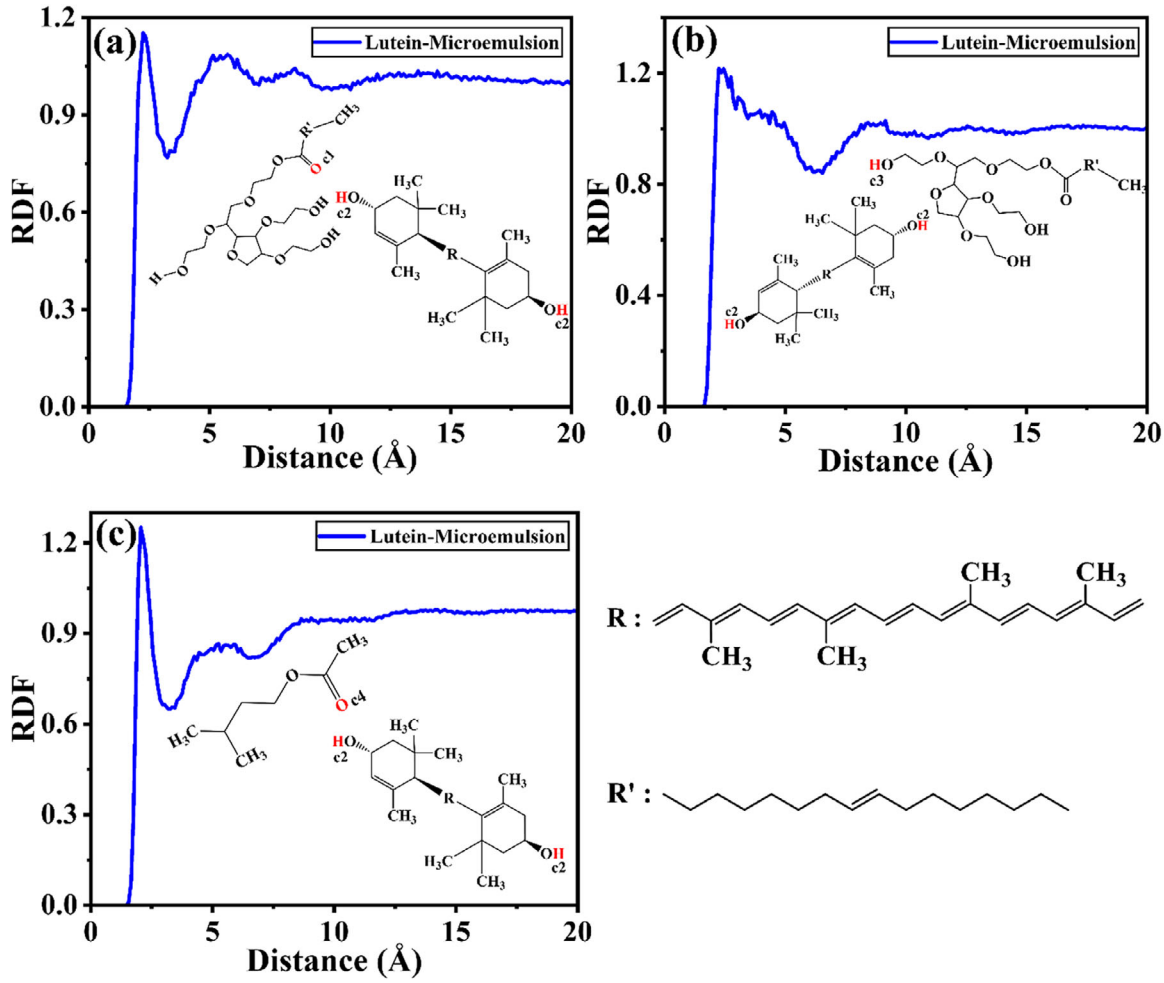


Figure 5. RDF graph of interaction between lutein and components of Microemulsion a) O (tween 80)- H (hydroxyl moiety of lutein), b) O (hydroxyl moiety of tween 80)- H (hydroxyl moiety of lutein), c) O (ester moiety of IAA)- H (hydroxyl moiety of lutein).

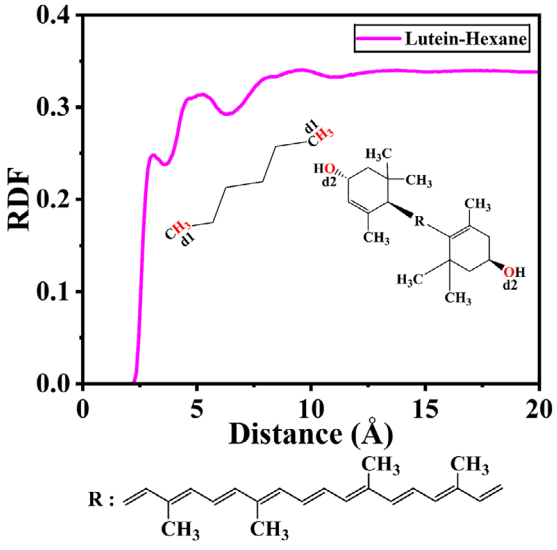


Figure 6. RDF graph of interaction between H (terminal alkyl moiety of hexane)- O (hydroxyl moiety of lutein).

the Lutein-2MeTHF, Lutein-DES, and Lutein-Microemulsion systems form H-bonds with lutein. Among them, the Lutein-DES system exhibits the highest H-bonding compared to other systems. However, the Lutein-hexane system does not form H-bonding due to the absence of electronegative atoms and the non-polar nature of hexane. The RDF and *CN* analysis indicate the first solvation shell in the H-bonding region for the Lutein-2-MeTHF, Lutein-DES, and Lutein-Microemulsion systems. However, in the Lutein-DES system, the RDF analysis indicates strong H-bonding within the DES (i.e., between HBA and HBD). However, weaker hydrogen bonds between lutein and DES components suggest competition for H-bonding sites. In Lutein-Microemulsion, RDF curves demonstrate varied interactions with microemulsion components. Thus, minimal interactions with the hydroxyl moiety of Tween80 and stronger interactions with the ester moiety of isoamyl acetate. The solvation shells in this system reflect diverse and complex solute-solvent dynamics. The MSD analysis indicates subdiffusive behavior of lutein in Lutein-2-MeTHF, Lutein-Microemulsion, and Lutein-Hexane systems, attributed to hindrance from solvent interactions. However, the Lutein-DES system with a lower β value reflects strong

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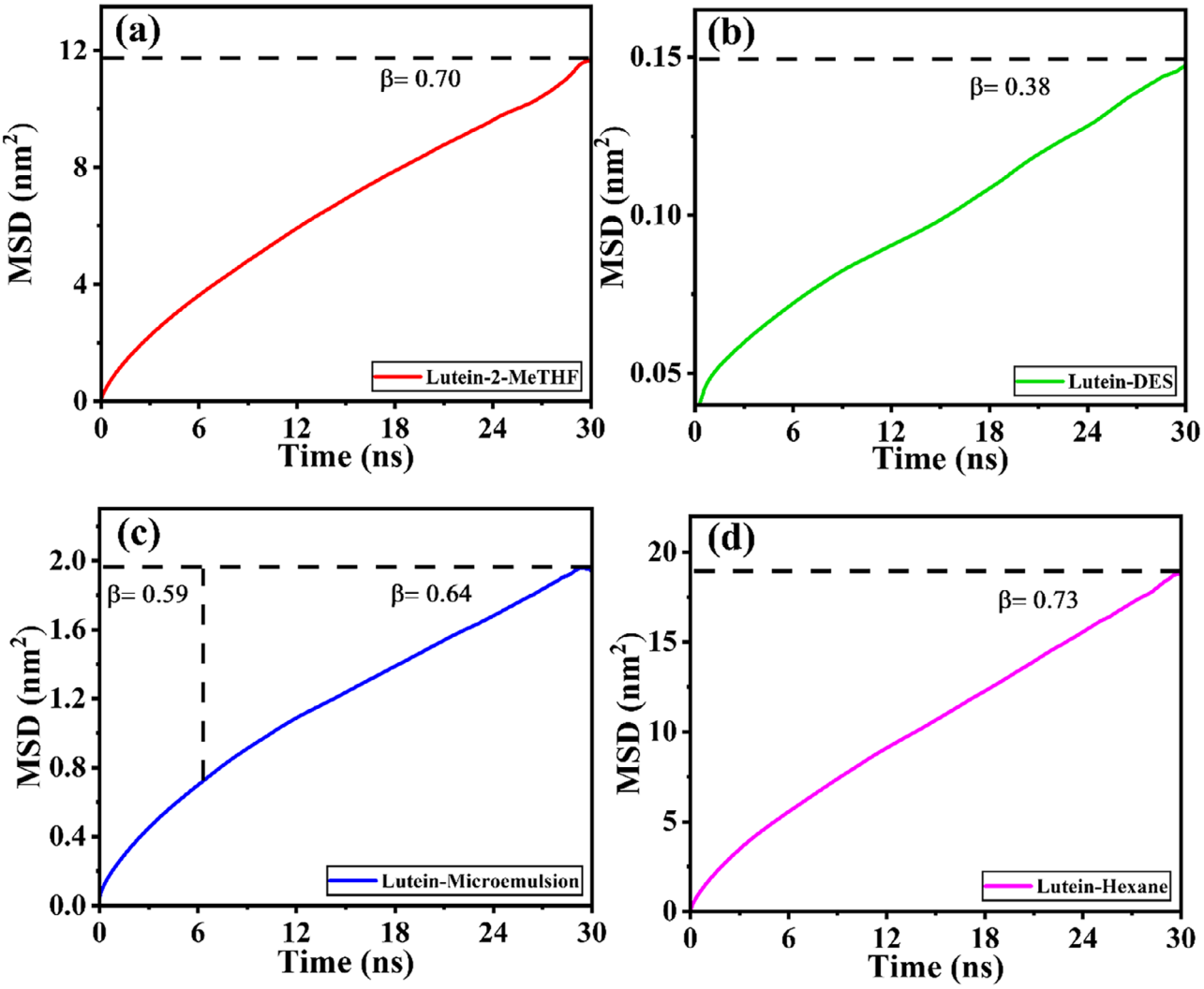


Figure 7. Represents the MSD curve of the Hydrogen atom of the hydroxyl moiety of lutein in different solvent systems a) Lutein-2-MeTHF, b) Lutein-DES, c) Lutein-Microemulsion, and d) Lutein-Hexane.

subdiffusive behavior, significantly slowing down the movement of lutein within a system. The analysis highlights that the Lutein-Hexane system exhibits minimal interaction strength, leading to less efficient extraction compared to more interactive solvents like 2-MeTHF, DES and microemulsions, and the lutein-DES system exhibits maximum interaction strength. This supports the conclusion that DES is more suitable for lutein extraction due to its better solvation capacity and interaction dynamics, which is suggested by the experimental literature. This atomistic scale interaction helps to understand the interaction between solute and solvents without performing experiments. This method offers a valuable alternative to traditional trial-and-error approaches commonly employed in laboratory-based extraction method development. In future studies, we will extend this simulation-based approach to other xanthophylls and structurally related bioactive compounds, aiming to develop broadly applicable and sustainable green extraction strategies.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

lutein, molecular dynamics simulation, solvent selection, transport property

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